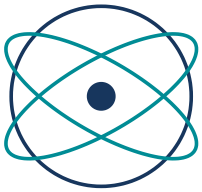
EXAM REVISION SERIES

PHYSICS FORMULAS & METHODOLOGIES

NCERT Class 11 Physics — Chapter 1



Subject: Physics (Class 11)

Chapter: Units and Measurement

Target Exams: CBSE / JEE Main / NEET

Standard: Scientific Navy & Teal

Document Features:

- ✓ Rigorous LaTeX Formula Typesetting
- ✓ Precise Step-by-Step Derivations
- ✓ Systematic Numerical Problem Frameworks
- ✓ Common Calculation Error Warnings

Strictly adheres to official NCERT Curriculum & SI (BIPM 2018) Standards

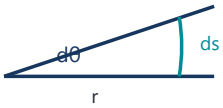
1. FUNDAMENTAL UNITS & ANGLES

Plane Angle & Solid Angle

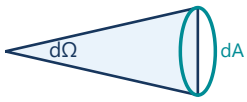
MUST KNOW

$$d\theta = \frac{ds}{r}\ (\text{rad}), \quad d\Omega = \frac{dA}{r^2}\ (\text{sr})$$

Symbols: ds : Arc length, r : Radius, dA : Spherical surface area.
SI Units: Radian (rad) for plane angle; Steradian (sr) for solid angle.
Conditions: Both quantities are geometrically dimensionless ($[M^0L^0T^0]$).



(a) Plane Angle $d\theta = \frac{ds}{r}$



(b) Solid Angle $d\Omega = \frac{dA}{r^2}$

Significant Figures Rules

Rule Condition	Example	Significant Digits
All non-zero digits are significant.	$2308\ \text{cm}$	4 $(2, 3, 0, 8)$
Zeros between two non-zero digits are significant.	$2.008\ \text{m}$	4 $(2, 0, 0, 8)$
Leading zeros (left of first non-zero digit) are NOT significant.	$0.0023\ \text{kg}$	2 $(2, 3)$
Trailing zeros in a number WITHOUT a decimal point are NOT significant.	$12300\ \text{m}$	3 $(1, 2, 3)$
Trailing zeros in a number WITH a decimal point ARE significant.	$3.500\ \text{g}$	4 $(3, 5, 0, 0)$

⚠ CAUTION: Unit Conversion Trap

Changing units does **NOT** alter the number of significant figures. Always express numbers in scientific notation ($a \times 10^b$) to avoid ambiguity.
Example: $4.700\ \text{m} = 4.700 \times 10^2\ \text{cm} = 4.700 \times 10^{-3}\ \text{km}$ (All retain 4 significant figures).

2. PROPAGATION OF ERRORS

Error Combination Rules

EXAM FAVOURITE

$$\text{Addition/Subtraction: } Z = A \pm B \implies \Delta Z = \Delta A + \Delta B$$

$$\text{Product/Division: } Z = A \cdot B \text{ or } \frac{A}{B} \implies \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$\text{Power Relation: } Z = A^p B^q / C^r \implies \frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$$

Absolute Error: ΔZ is always additive to compute maximum possible error.

Percentage Error: $\%$ Error in $Z = \left(\frac{\Delta Z}{Z}\right) \times 100\%$.

Systematic Numerical-Solving Framework

Methodology: Calculating Error in Derived Physical Quantities

1. **State Given Measurements:** $Length\ (l) = 16.2 \pm 0.1$ cm, $Breadth\ (b) = 10.1 \pm 0.1$ cm.
2. **Compute Measured Area:** $A = l \times b = 16.2 \times 10.1 = 163.62\ \text{cm}^2$.
3. **Apply Fractional Error Rule:** $\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b} = \frac{0.1}{16.2} + \frac{0.1}{10.1} \approx 0.00617 + 0.00990 = 0.01607$.
4. **Calculate Absolute Uncertainty:** $\Delta A = A \times 0.01607 = 163.62 \times 0.01607 = 2.63\ \text{cm}^2 \approx 3\ \text{cm}^2$.
5. **Final Reported Result:** $\mathbf{A = (164 \pm 3)\ cm^2}$.

Core Dimensional Formulas

Physical Quantity	Defining Equation	Dimensional Formula	SI Unit
Density (ρ)	$\rho = \frac{\text{Mass}}{\text{Volume}}$	$[M^1L^{-3}T^0]$	kg m^{-3}
Force (F)	$F = m \cdot a$	$[M^1L^1T^{-2}]$	$\text{N}\ (\text{kg m s}^{-2})$
Work / Energy (E)	$W = F \cdot s$	$[M^1L^2T^{-2}]$	$\text{J}\ (\text{kg m}^2\text{s}^{-2})$
Pressure / Stress (P)	$P = \frac{\text{Force}}{\text{Area}}$	$[M^1L^{-1}T^{-2}]$	$\text{Pa}\ (\text{N m}^{-2})$

3. DIMENSIONAL ANALYSIS & APPLICATIONS

Principle of Homogeneity of Dimensions

MUST KNOW

$$\text{If } A + B = C \implies [A] = [B] = [C]$$

Rule: Physical quantities can only be added or subtracted if they possess the same dimensions.

Note: Arguments of exponential, trigonometric, and logarithmic functions must be dimensionless.

Rigorous Derivation Method

Derivation of Simple Pendulum Time Period (T) via Dimensional Analysis

1. **Assume Product Dependence:** $T = k \cdot l^x \cdot g^y \cdot m^z$, where k is a dimensionless constant.
2. **Substitute Dimensional Formulas:** $[M^0L^0T^1] = [M^0L^1T^0]^x \cdot [M^0L^1T^{-2}]^y \cdot [M^1L^0T^0]^z$.
3. **Collect Exponents:** $[M^0L^0T^1] = M^z \cdot L^{x+y} \cdot T^{-2y}$.
4. **Equate Exponents on Both Sides:**
   - Mass exponent: $z = 0$
   - Time exponent: $-2y = 1 \implies y = -\frac{1}{2}$
   - Length exponent: $x + y = 0 \implies x = -y = \frac{1}{2}$
5. **Substitute Exponents Back:** $T = k \cdot l^{1/2} \cdot g^{-1/2} \cdot m^0 \implies \mathbf{T = k\sqrt{\frac{l}{g}}}$.
6. **Final Formula (Experimentally $k = 2\pi$):** $\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$.

⚠ Limitations of Dimensional Analysis

1. Cannot determine dimensionless constants (k, 2π, $\sin\theta$).
2. Fails for expressions containing sum/difference of terms (e.g., $s = ut + \frac{1}{2}at^2$).
3. Cannot derive formulas with more than 3 independent physical variables.